

breed of high-density, flexible servers for accelerated computing.

The second approach is based on the disaggregation of data centre resources where the individual compute, memory and storage resources are connected via the network fabric and can be individually optimised and scaled in line with the cloud paradigm.

## CAPI on Power9

CAPI on Power9 system provides a high-performance solution for implementation of client-specific, computation-heavy algorithms on an FPGA. It can replace either application programs running on a core or custom acceleration implementations attached via input/output (I/O). Applications for CAPI include Monte Carlo algorithms, key value stores and financial and medical algorithms. CAPI removes the overhead and complexity of the I/O subsystem, allowing an accelerator to operate as part of an application.

A specific algorithm for acceleration is contained in a unit on the FPGA called the accelerator function unit

(AFU or accelerator). AFU provides applications with a higher computational unit density for customised functions to improve the performance of the application and offload the host processor. Use of AFU allows cost-effective processing over a wide range of applications. CAPI can also be used as a base for flash memory expansion, as is the case for

IBM Data Engine for NoSQL Power Systems Edition.

The Coherent accelerator processor proxy (CAPP) unit maintains a directory of all cache lines held by the off-chip accelerator, allowing it to act as the proxy that maintains architectural coherence for the accelerator across its virtual memory space. A power service layer (PSL) resides on

the FPGA alongside the acceleration engine. It works in concert with the CAPP unit across a PCIe connection. The PSL provides a straightforward command to the client-written accelerator, which grants access to coherent memory.

The CAPP and PSL handle all virtual-to-physical memory translations, simplifying the programming model and freeing the accelerator to do the number crunching directly on the data it receives. In addition, the PSL contains a 256kB resident cache on behalf of the accelerator. Based on the needs of the algorithm, the accelerator can direct the use of the cache via the type of memory accesses (read/write) as cacheable or non-cacheable. While

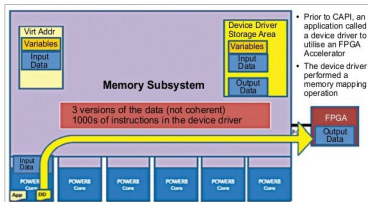


Fig. 10: Issues with PCI accelerators

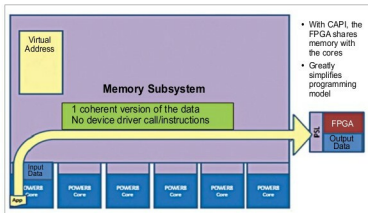


Fig. 11: Benefits of coherent accelerators

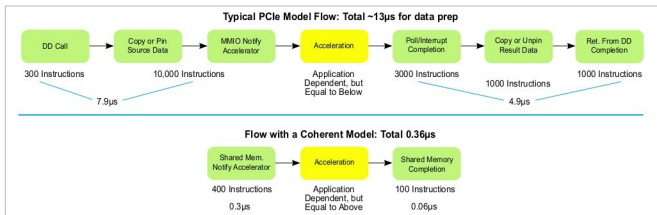


Fig. 12: Typical PCIe model flow